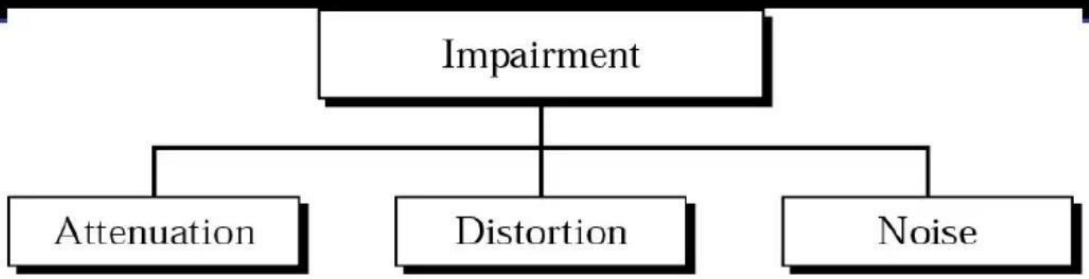
Transmission Impairment

- Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
  - **What is sent is not what is received.**
- Three causes of impairment are:
  - Attenuation
  - Distortion
  - Noise



Transmission Impairment

Attenuation

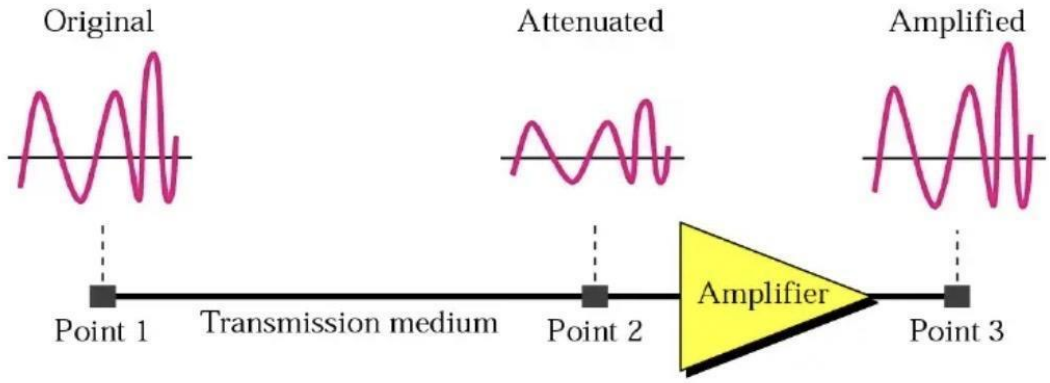
- loss of signal strength with distance over any transmission medium.
  - For guided media, attenuation, is generally exponential and thus is typically expressed as a constant number of decibels per unit distance. ***Due to resistance***
  - For unguided media, attenuation is a more complex function of distance and the makeup of the atmosphere.
- To compensate for this loss, amplifiers are used to amplify the signal.
- Attenuation varies with frequency.

Transmission Impairment

Impairment types

Attenuation

- Figure below shows the effect of attenuation and amplification.



Transmission Impairment

Attenuation

Decibel

- To show that a signal has lost or gained strength, we use the unit of the decibel.
- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.
- Note that the decibel is:
  - Negative if a signal is attenuated and
  - Positive if a signal is amplified.

$$\mathbf{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

- PI and P2 are the powers of a signal at points 1 and 2, respectively.

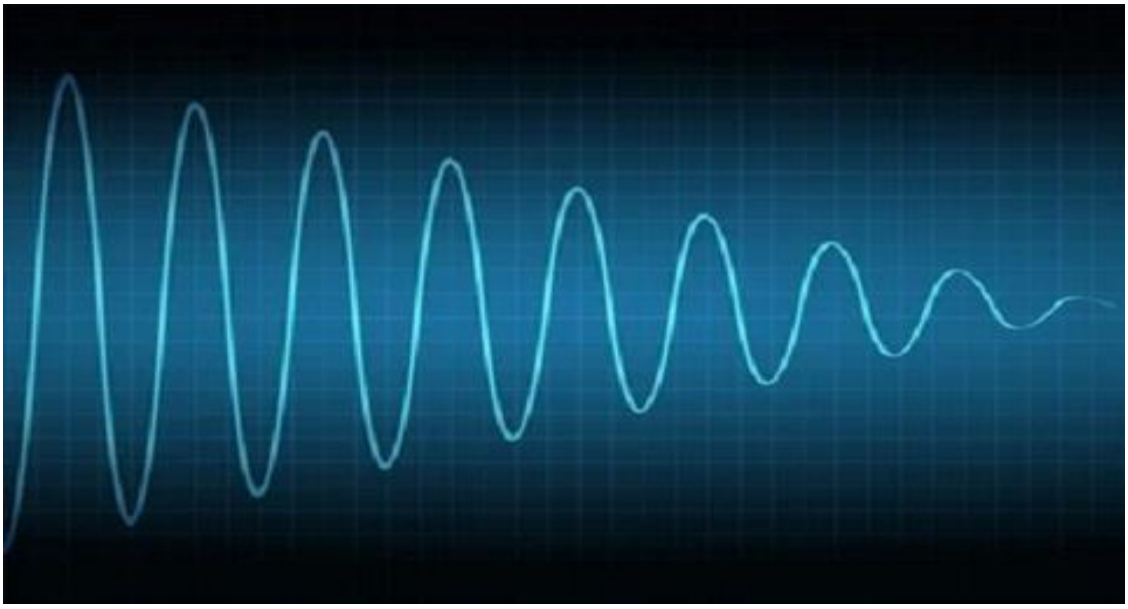


Fig:Attenuation

Transmission Impairment

Attenuation

Example

- Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P2 = ½ P1.
- In this case, the attenuation (loss of power) can be calculated as:

$$
\begin{aligned}
dB &= 10 \log_{10} (P2 / P1) \\
&= 10 \log_{10} (1/2\ P1 / P1) \\
&= 10 \log_{10} (0.5)) \\
&= 10 (-0.3) \\
dB &= -3
\end{aligned}
$$

Transmission Impairment

Decibel value will be Positive if a signal is amplified.

Example

EXAMPLE 3.8 Decibels are useful in determining the gain or loss over a series of transmission elements. Consider a series in which the input is at a power level of 4 mW, the first element is a transmission line with a 12-dB loss (−12-dB gain), the second element is an amplifier with a 35-dB gain, and the third element is a transmission line with a 10-dB loss. The net gain is $(-12 + 35 - 10) = 13$ dB. To calculate the output power P_{out}:

$$G_{dB} = 13 = 10 \log(P_{out}/4 \text{ mW})$$
$$P_{out} = 4 \times 10^{1.3} \text{ mW} = 79.8 \text{ mW}$$

Transmission Impairment

Distortion

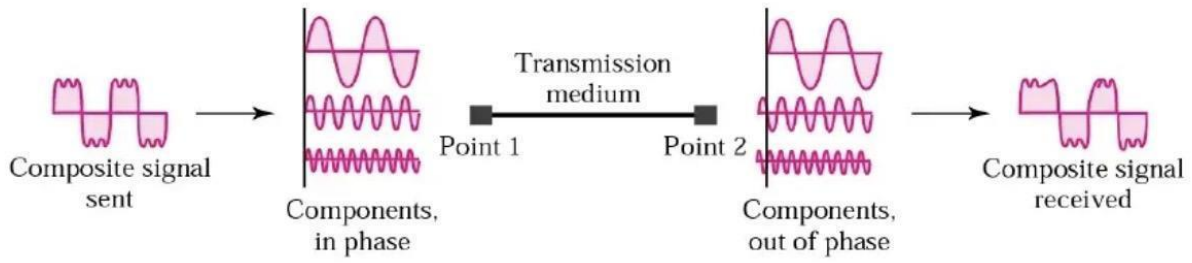
- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
  - Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.
  - Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.
  - In other words, signal components at the receiver have phases different from what they had at the sender.
  - The shape of the composite signal is therefore not the same.

Due to propagation speeds

Transmission Impairment

Distortion

- Figure below shows the effect of distortion on a composite signal.



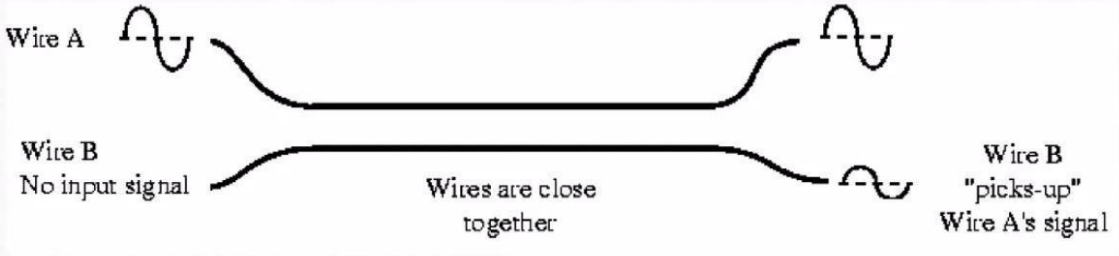
Noise

- **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
  - Thermal noise is uniformly distributed across the systems and hence is often referred to as **white noise**, cannot be eliminated.

- **Intermodulation noise** When signals at different frequencies share the same transmission medium.
  - It produce signals at a frequency that is the sum or difference of the two original frequencies or multiples of those frequencies

- Cross-talk it is an unwanted coupling between signal paths. It can occur by electrical coupling between nearby twisted pairs.
- Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on. It has much effect on digital data.

Cross talk



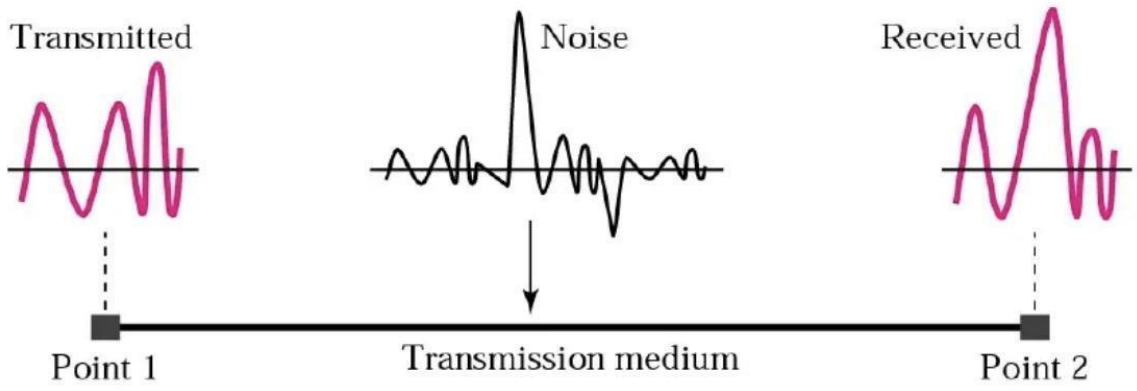
Cross talk

- Data signals may also be subjected to something commonly referred to as crosstalk, which occurs when signals from two cables in close proximity to one another interfere with each other.
- As a result, the signals on both cables may become corrupted.
- Cross talk occurs in guided media .

Transmission Impairment

Noise

- Figure shows the effect of noise on a signal.



Transmission Impairment

Noise

- Signal-to-Noise Ratio (SNR)

- To find the theoretical bit rate limit, we need to know the ratio of the signal power to the noise power.
- The signal-to-noise ratio is defined as:

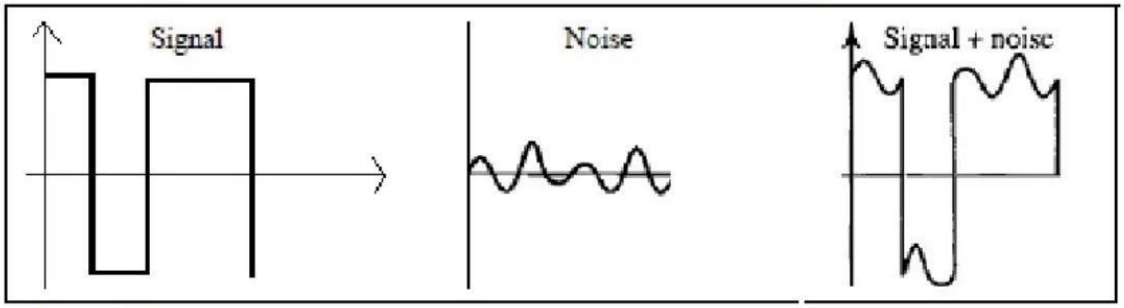
  SNR = average signal power / average noise power

- We need to consider the average signal power and the average noise power because these may change with time.

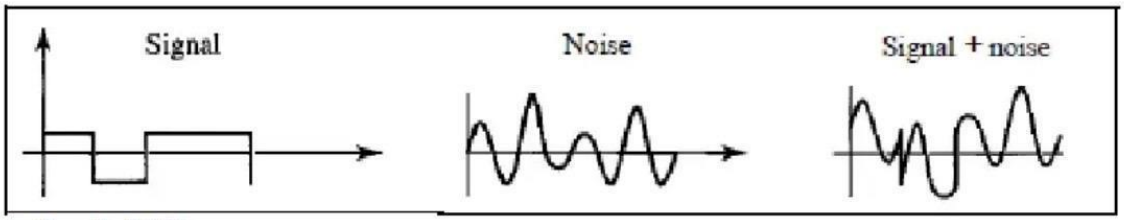
Transmission Impairment

Noise

- Figure shows the idea of SNR



a. Large SNR



b. Small SNR

Transmission Impairment

Noise

- Signal-to-Noise Ratio (SNR)

- SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
  - A high SNR means the signal is less corrupted by noise;
  - A low SNR means the signal is more corrupted by noise.

- Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB}, defined as:

$$SNR_{dB} = 10 \log_{10} SNR$$

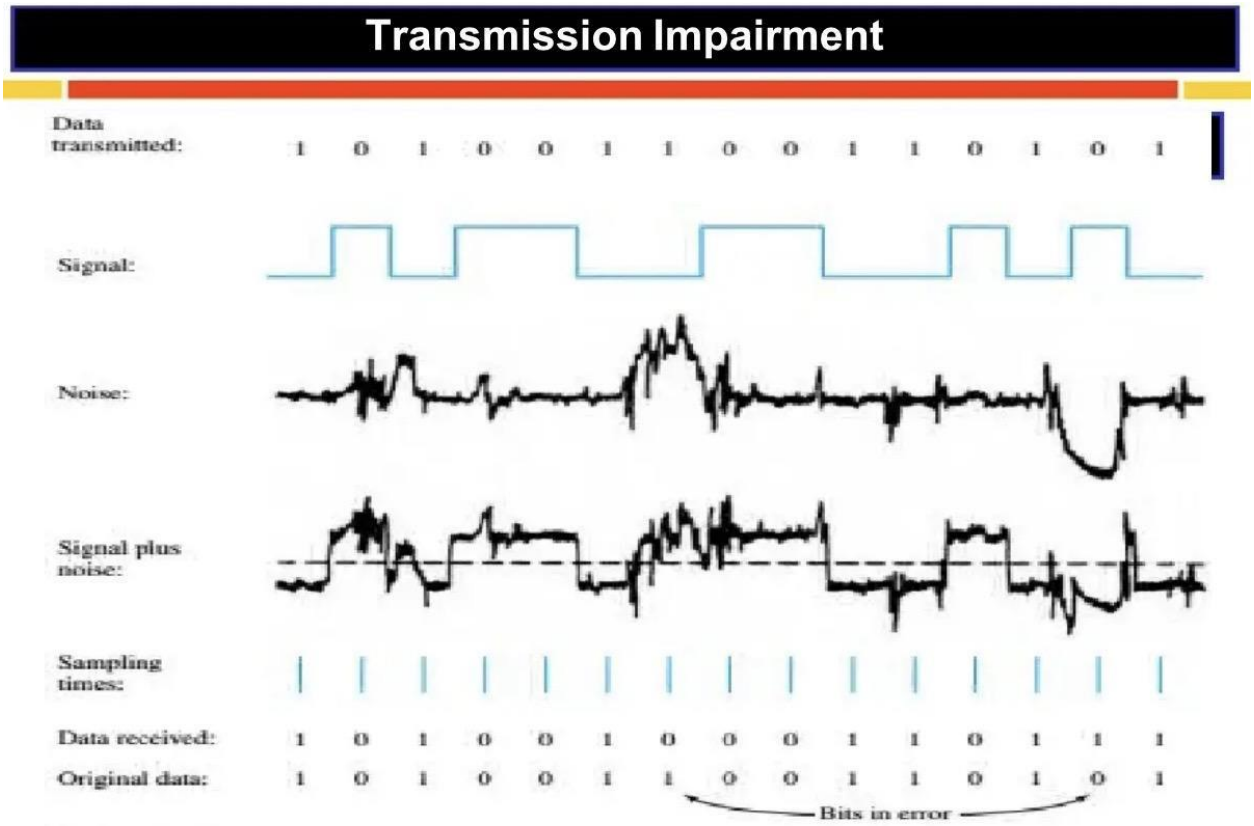


Figure 3.16 Effect of Noise on a Digital Signal

Classification of Noise:

There are several way to classify Noise, but conveniently Noise is classified as

1) External Noise

2) Internal Noise

External Noise:

External noise is defined as the type of Noise which is general externally due to communication system. External Noise are analysed qualitatively. Now, External Noise may be classified as

a) Atmospheric Noise : Atmospheric Noise is also known as static noise which is the natural source of disturbance caused by lightning, discharge in thunderstorms and the natural disturbances occurring in nature.

b) Industrial Noise : Sources of Industrial noise are auto-mobiles, aircraft, ignition of electric motors and switching gear. The main cause of Industrial noise is High voltage wires. These noises are generally produced by the discharge present in the operations.

c) Extraterrestrial Noise : Extraterrestrial Noise exists on the basis of their originating source. They are subdivided into

i) Solar Noise

ii) Cosmic Noise

Internal Noise:

Internal Noise is the type of Noise which is generated internally or within the Communication System or in the receiver. They may be treated qualitatively and can also be reduced or minimized by the proper designing of the system. Internal Noises are classified as

1) Shot Noise : These noises generally arise in the active devices due to the random behaviour of Charge particles or carries. In the case of an electron tube, shot noise is produced due to the random emission of electron form cathodes.

2) Low- Frequency Noise : They are also known as FLICKER NOISE. These type of noise are generally observed at a frequency range below a few kHz. Power spectral density of these noise increases with the decrease in frequency. That is why the name is given Low- Frequency Noise.

3) Thermal Noise : Thermal Noise is random and often referred as White Noise or Johnson Noise. Thermal noise is generally observed in the resistor or the sensitive resistive components of a complex impedance due to the random and rapid movement of molecules or atoms or electrons.

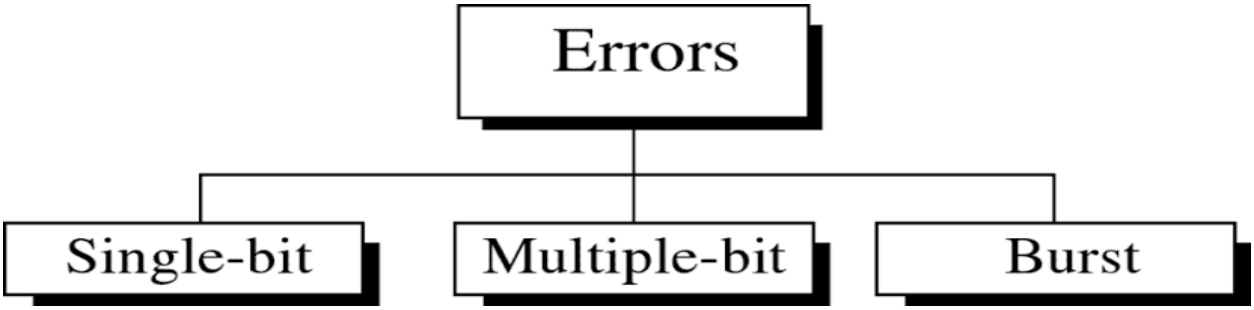
ERROR

1. Data are transmitted in the network.
2. The data can be corrupted during transmission.
3. Transmission error.
4. For reliable communication, errors must be detected and corrected

5. Error detection and correction are implemented either at the data link layer or the transport layer of the OSI model.

Error types

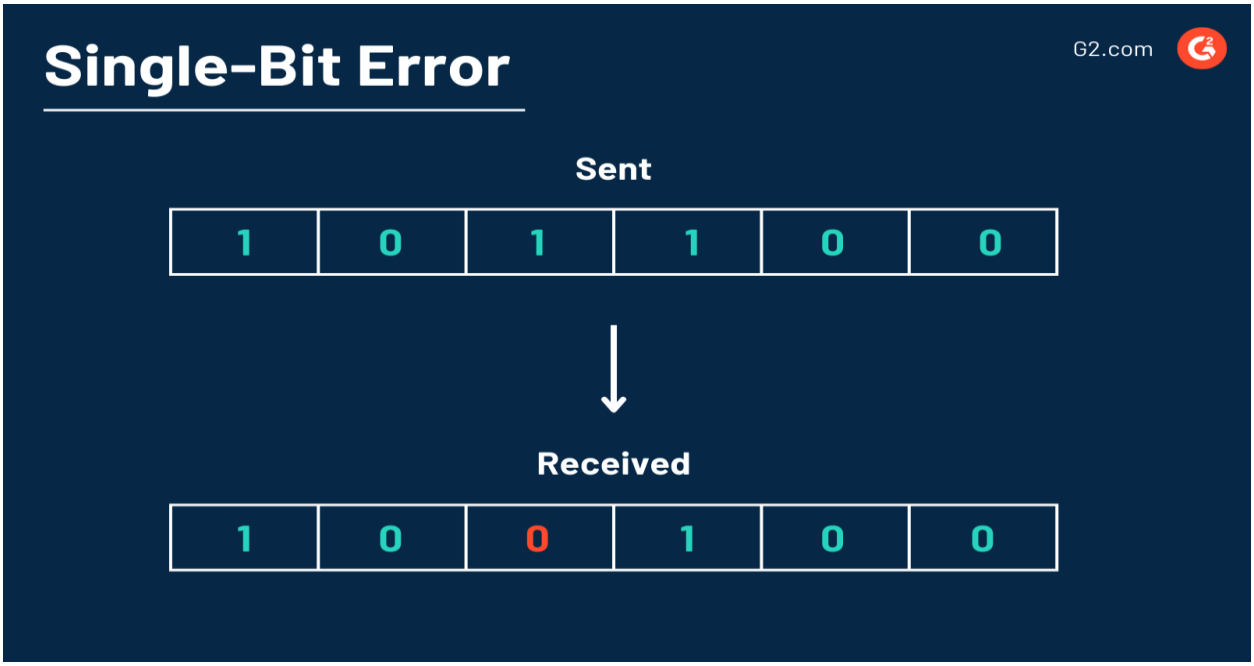
In data transmission, the primary **error types** concerning the number of corrupted bits are **single-bit errors** and **multiple-bit errors**, the later of which can be further categorized as **burst errors**.



These errors occur due to noise and interference changing the values of bits from 0 to 1 or 1 to 0 during transmission.

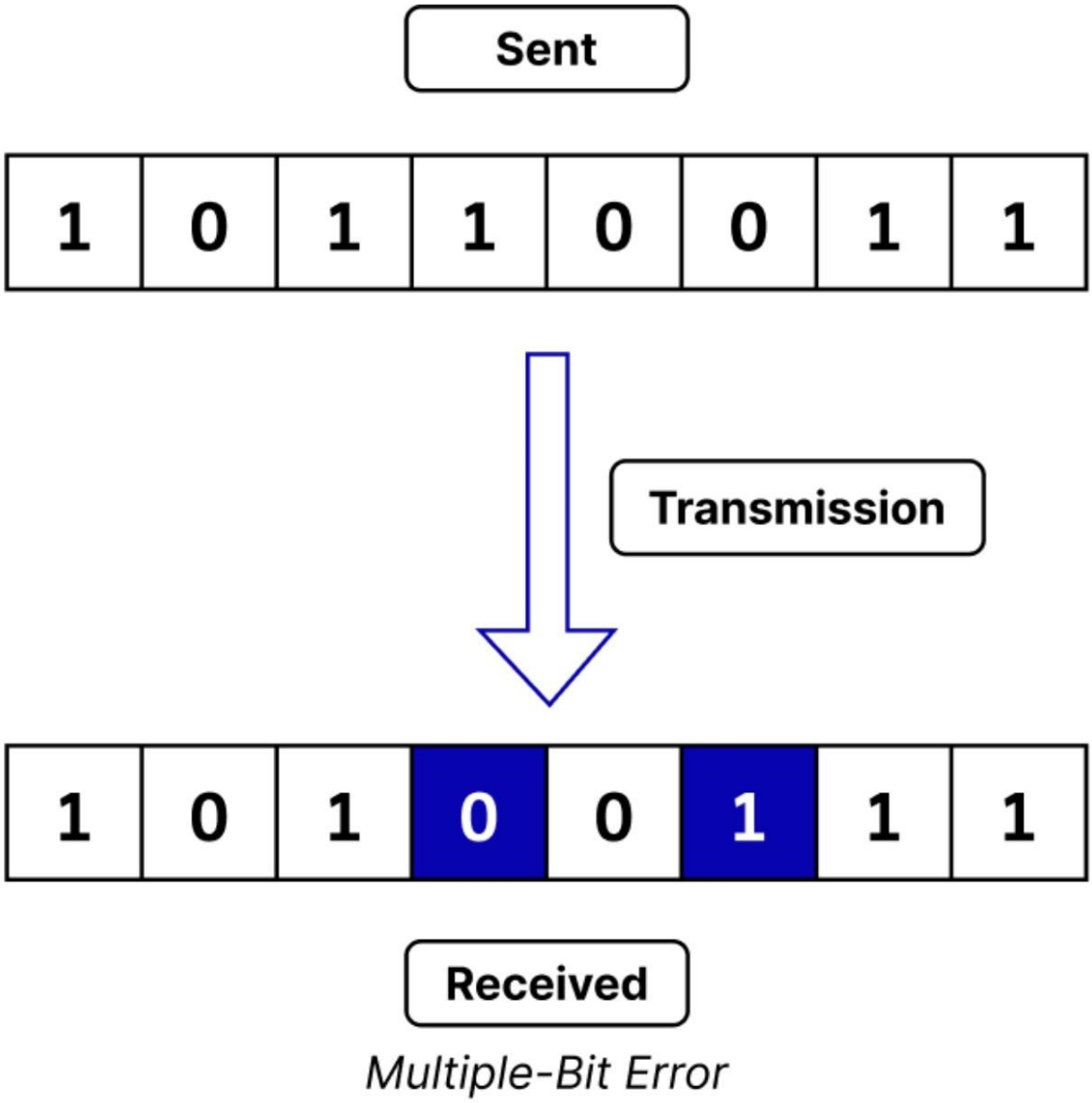
- ## Single-Bit Error

A single-bit error occurs when only one bit within a given data unit (such as a byte or frame) is altered during transmission.



- **Characteristics**: Only one bit anywhere in the data unit is flipped (0 becomes 1, or 1 becomes 0).
- **Occurrence**: These are less likely in high-speed serial data transmission where each bit lasts for a very short duration, but can occur in parallel transmission if one of the wires is noisy.
- ## Multiple-Bit Error

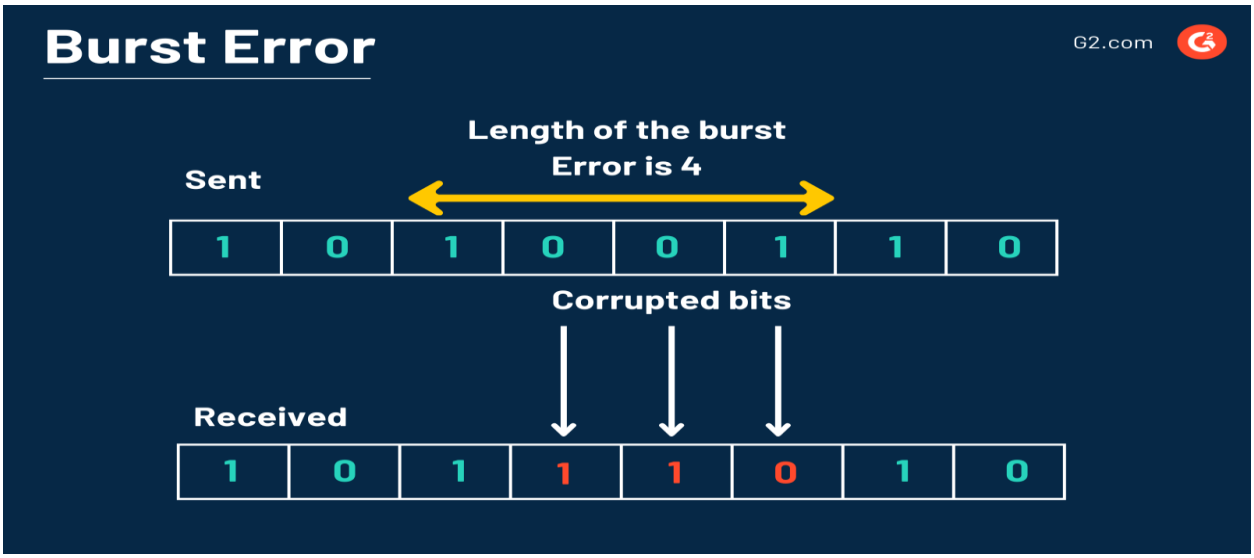
A multiple-bit error occurs when more than one bit in a data unit is corrupted. The corrupted bits are not necessarily consecutive.

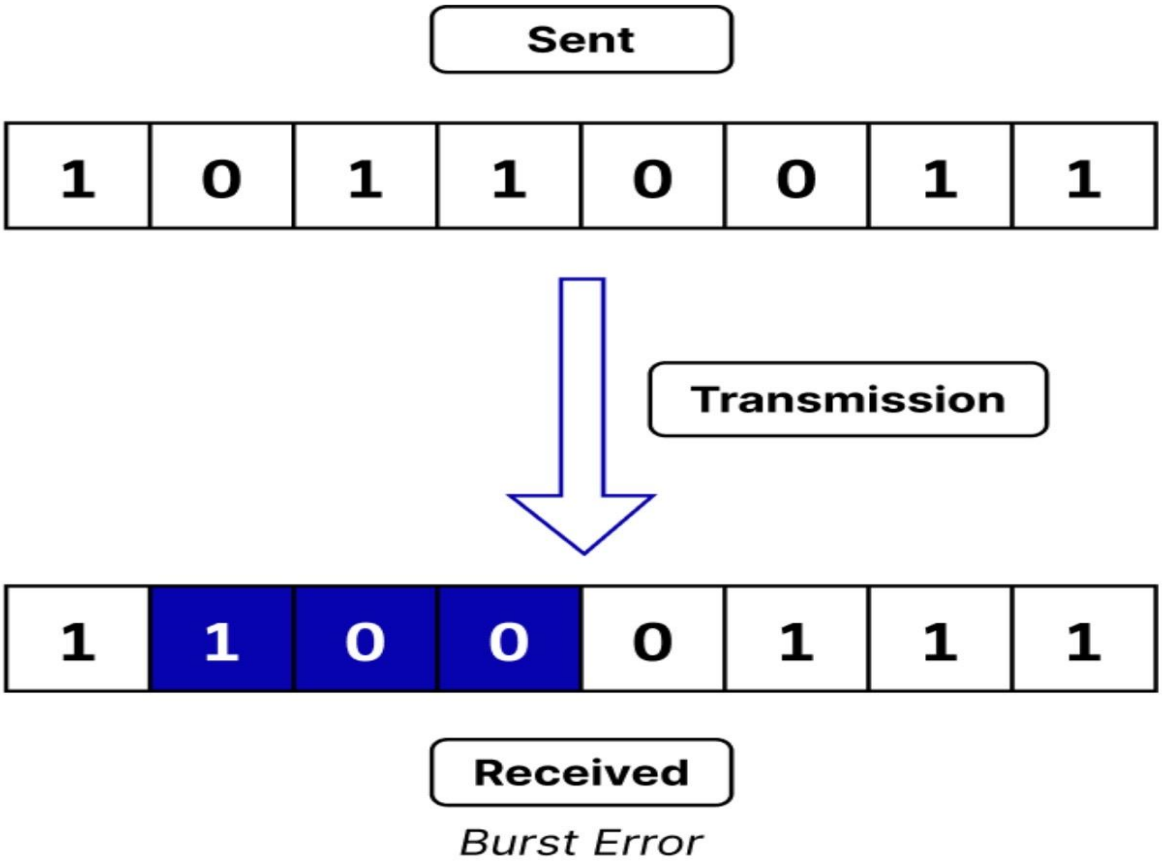


- **Characteristics**: Two or more bits are affected within the same data unit.
- **Occurrence**: These are more likely to occur in high-noise environments.

- **Burst Error (Subtype of Multiple-Bit Error)**

A burst error is a specific type of multiple-bit error where two or more corrupted bits are consecutive or occur within a contiguous sequence.





- **Characteristics**: The length of the burst error is measured from the first corrupted bit to the last corrupted bit, potentially including some correct bits in between.

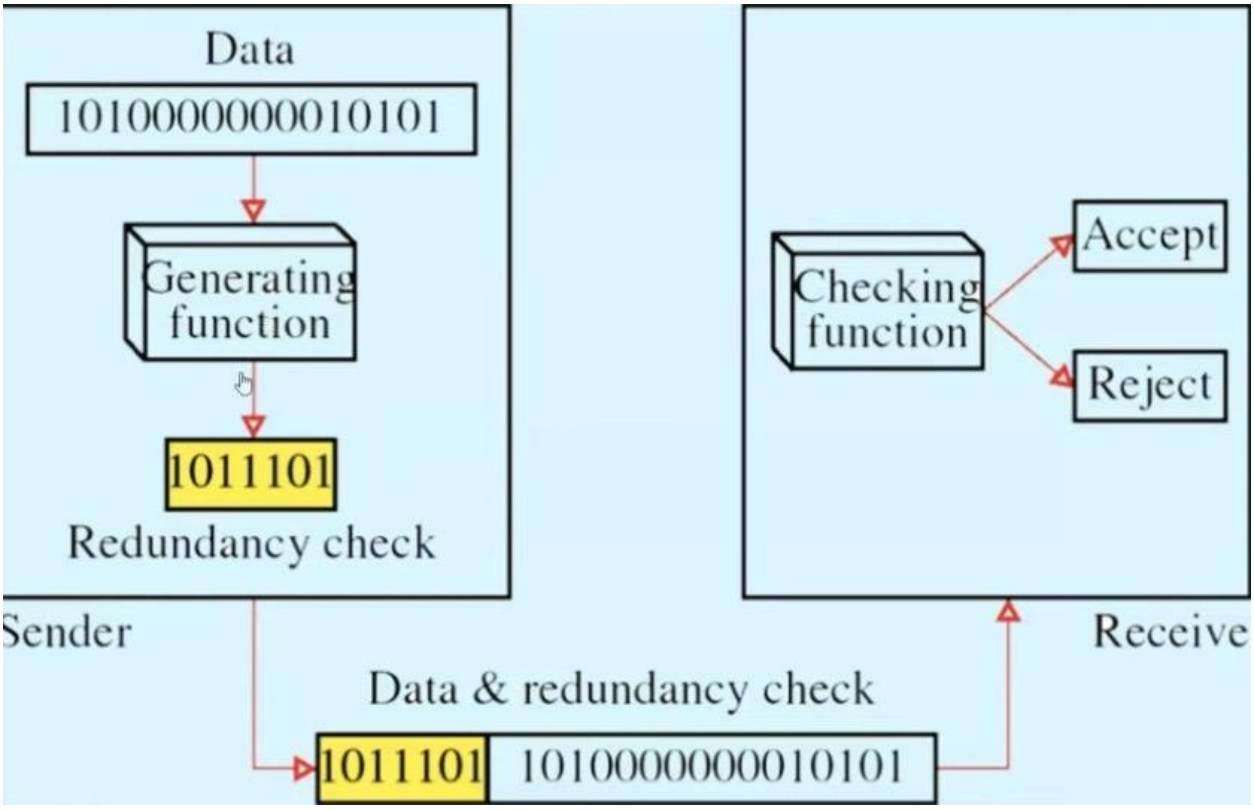
- **Occurrence**: Burst errors are common in serial data transmission because a single, longer duration of noise can affect a whole sequence of bits.

HOW TO DETECT THE ERRORS?

Error detection means to decide whether the received data is correct or not without having a copy of the original message.

- To detect or correct errors, we need to send some extra bits with the data.
- The extra bits are called as redundant bits.

Redundancy

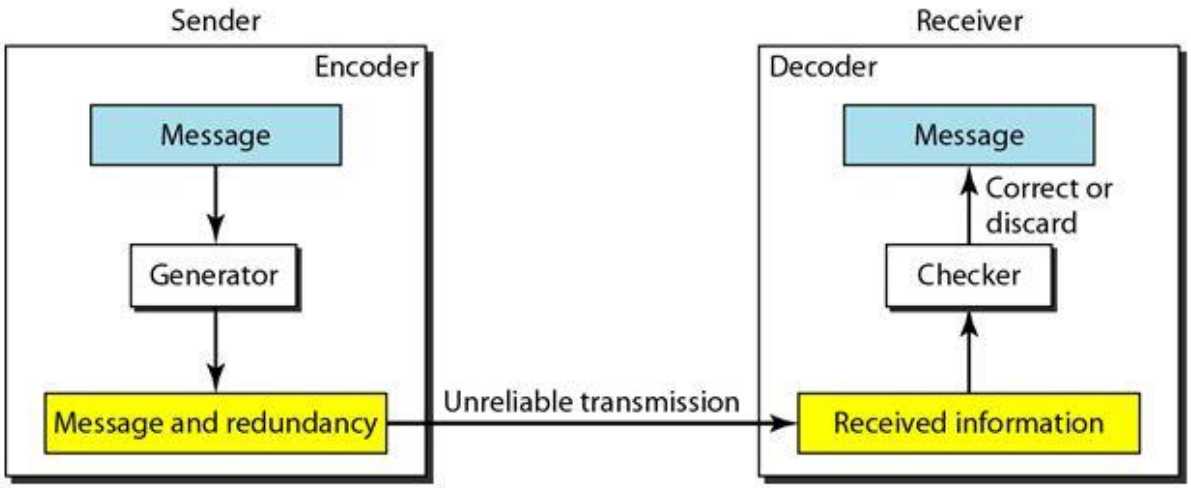


ERROR CORRECTION

It can be handled in two ways:

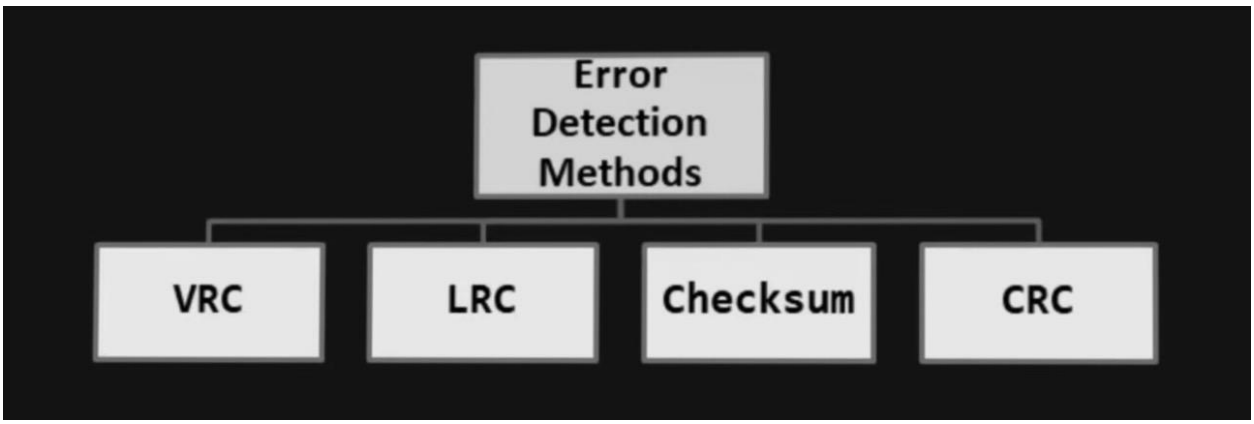
- Receiver can have the sender retransmit the entire data unit.
- The receiver can use an error-correcting code, which automatically corrects certain errors.

Error Detection /Correction



ERROR DETECTION TECHNIQUES

Four types of redundancy checks are used in data communications. They are:

- Vertical Redundancy Check (VRC)
- Longitudinal Redundancy Check (LRC)
- Checksum
- Cyclic Redundancy Check (CRC)

Error Detection